Garimella Abhimanyu Koushik

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	9.64
FIITJEE Dilshuknagar	Class XII (Telangana State Board of Intermediate Education)	2024	96.00%
Defence Laboratories school	Class X (Central Board of Secondary Education)	2022	98.20%

SCHOLASTIC ACHIEVEMENTS

Academic Excellence Award: Highest CGPA in the **Department of EE** for the year 2025-2026 at IIT Hyderabad.

A+ grades: Computer Architecture, Probability, and Random Processes.

SKILLS

Programming: C, C++, Python, RISCV Assembly

HDL: Verilog, SystemVerilog

Tools and EDA: Vivado, Intel Quartus, Make, CMake, Git, GDB, Linux, LTSpice

MAJOR PROJECTS

Custom Chess Engine | C++ | Github

- Developed a high-performance, **UCI-compliant chess engine** (**~2000 ELO**) from scratch in **C++20** under strict memory and compute constraints.
- Implemented bitboard-based move generation with **Magic Bitboards**, **iterative deepening**, **alpha-beta search**, Zobrist-hashed **transposition tables**, and **quiescence search** for tactical stability.
- Optimized move ordering with **MVV-LVA** and history heuristics, and applied **tapered piece-square tables** for evaluation.
- Architected a **multi-threaded Lazy SMP parallel search** framework with **lockless shared-memory access** to a global transposition table and **custom bit-packing** for cache-line density.
- Mitigated thread race conditions using **atomic operations**, ensuring thread safety during deep-search state exploration.
- Validated structural correctness using custom **C++ test benches** via **GDB**.

64-bit RISC-V Processor Core | SystemVerilog | Github

- Designed a **64-bit RISC-V CPU** with a **5-stage pipeline** featuring a **hazard detection unit** and full **data forwarding**.
- Built a 64-entry **dynamic branch predictor** with IP-indexed history tracking (taken status + target address).
- **Integrated direct-mapped L1 instruction** and **data caches** with a single-port main memory via a **custom memory arbiter**, modeling 100-cycle memory latency with global pipeline freeze/stall signals.
- Developed a structural SystemVerilog **testbench framework** for RTL-level hardware verification, covering hazard collisions, register file bypasses, and branch recovery.

Streaming R2SDF FFT Accelerator | Verilog | Github

- Designed and synthesized a streaming **Radix-2 Single-Delay Feedback** (R2SDF) FFT hardware accelerator to process continuous signal **data streams** on-chip.
- Configured an **11-bit (Q6.4)** resolution data path, integrated automated **Python**-generated combinatorial **twiddle**

factor ROMs, exploited structural coefficient symmetry to fold butterfly delays, and implemented an **output reorder buffer** to correct bit-reversed frequency indices.
- Successfully deployed the accelerator onto an **Intel Cyclone V** (DE10-Nano) FPGA using only **654 ALMs and 9 DSP blocks**, and built a Python framework to parse Signal Tap logic analyzer hex files to isolate and plot frequency spectrum magnitude bins.

MINOR PROJECTS

- **MNIST Digit Classification from Scratch | Github:** Built a fully connected **deep neural network** for handwritten digit recognition from **first principles** using **NumPy**. Manually wrote code for forward propagation, cross-entropy loss, and **backpropagation via gradient descent** without external ML frameworks.

- **FIR Filter | Github:** Implemented a Finite Impulse Response digital filter in **Verilog** and implemented it on **Intel Cyclone V FPGA**. Optimized the data path using **fixed-point representations** and balanced pipeline registers to minimize multiplication propagation delay and maximize operating frequency.

- **Multiplication and Division Hardware Unit | Github:** Engineered **parameterizable fixed-point** multiplier and divider hardware modules in **Verilog** by implementing a 4-phase arithmetic pipeline (Normalization, Seed Generation, Iterative Refinement, and Denormalization) utilizing the **Newton-Raphson** method. Refined the initial guess using an internal case-statement **LUT based on localized piecewise constant approximations** that slashed worst-case initial guess error.

- **Huffman Encoding | Github:** Developed a **lossless data compression** engine in **C++** using **priority queues** and **binary trees** to generate prefix codes. Implemented **bitwise shifting** to pack variable-length codes into byte-aligned memory streams.

- **Eigenvalue Calculator | Github:** Engineered a **complex matrix** eigenvalue calculator in **C** by using **Givens rotations** to execute **QR decomposition** and applying **Rayleigh Quotient Iteration** for better convergence rates.

RELEVANT COURSES

Computer Architecture, Data Structures, Digital Systems, Machine Learning, Signal Processing, Probability, Random Processes

POSITIONS OF RESPONSIBILITY

Core Member of **Elektronika** (Electronics and Signal Processing Club, IITH)**:** Designing an end-to-end **hardware Tetris game** in **Verilog** as part of a **4-member team**.

Teaching Assistant: Conducted tutorials, evaluated programming assignments, and held viva examinations for **Probability** and **Matrix Theory** courses.

EXTRACURRICULARS

Represented IIT Hyderabad at the **Inter IIT Chess** Championship (2024-2025).